Damping, Q, and Frequency Width

So far, we've talked about ideal resonators that don't lose energy. There's (unfortunately!) no such thing as a perpetual motion machine, so all resonators have to lose energy eventually. In this note we'll look at how that happens and what some of the practical implications are.

1. Damped Oscillators

The simplest model for the damping of an oscillator is an extra force that opposes motion. If you're moving to the right, then you feel a force to the left. If you're moving to the left, you feel a force to the right. Since the force is always in the opposite direction from your motion – the wind is always in your face – you slow down. This force is *resisting* the motion of the string. Mathematically, our usual wave equation picks up an extra term

$$\frac{\partial^2 y}{\partial t^2} = -b\frac{\partial y}{\partial t} + c_s^2\frac{\partial^2 y}{\partial x^2} \tag{1}$$

In Equation 1, we're really just saying that the force picks up an extra contribution. It used to be set by the bend of the string $\frac{\partial^2 y}{\partial x^2}$ but now we add in something that's opposed to the velocity. The velocity is the change of height with time, or $\frac{\partial y}{\partial t}$.

Before we added the resistance in, the modes of the string were

$$y(t,x) = \sin(kx)\sin(\omega t) \tag{2}$$

where k had to be picked so the string was zero at the ends, and we could work out ω because $c_s = \frac{\omega}{k}$. When we add in resistance, the new modes still look like sine waves along the string, but now decay away with time:

$$y(t,x) = \sin(kx)\sin(\omega t)\exp(-\frac{t}{\tau}) \tag{3}$$

I've plotted the time part of this in Desmos here. You should think of this as the height of one particular point on a string with time. We still have a sine wave, but it's going down with time/as we move to the right. The constant τ is the *decay time*, and is the length of time it takes for our signal to decay by a factor $1/e \simeq .367....$

I've made a Desmos plot here that shows what a damped string might look like. We start with a large-amplitude sine wave, but as time goes on, the amplitude decays away. The initial amplitude is 3, and $\tau = 10$. As the time goes to 10, the amplitude goes down by $1/e$, and the new amplitude is very close to 1. As the time increases to $t = 20$, we're down by another factor of $1/e$, and the wave is much smaller. If τ gets bigger, it takes longer for our string to damp down. If τ gets smaller, then we damp quickly.

One place you've probably experienced this is in a cathedral or concert hall, where we call τ the *reverb time*. The stone walls of a cathedral are very good at reflecting sound. If we make a sound, it will echo for a long time. In a cathedral, then, τ is very large. That's great for hymns, but bad for fast music. If notes haven't died away when we're playing the next ones, the sounds

overlap and music gets muddied. If τ is very small, then our sound goes away very quickly and we get very thin-sounding music. Concert halls usually target about 2 seconds for the reverb time; that gives you a full sound without getting too muddy.

2. Quality Factor

When we look at damped oscillations, a fundamental property is how *many* oscillations it takes for our signal to decay. This number is known as the *quality factor* or Q, and it plays a major role in how instruments (and concert halls!) behave. If our string vibrates for a lot of periods, we say it's high-Q. If it dies down after only a few periods, we say it's low-Q. Musical instruments tend to have Q's of several hundred.

Let's first look at how Q affects how loud an instrument can get. Let's say I have a motor at the end of a string that jiggles the end by 1mm in a sine wave. If I tune the motor to run at the string's fundamental frequency, then each of my 1mm jiggles will add in phase, and the string will move by a lot. How much is a lot? If the string's motion dies down after say 10 oscillations, so Q is 10, then I can add 10 of those 1mm jiggles together before the first ones fade away. That means my 1mm oscillations will turn into waves with an amplitude of 10mm/1cm. That's much bigger! If our Q goes up, then we can amplify our signal even more. As a rule of thumb, an oscillator "remembers" its last Q oscillations. If we're driving that oscillator at a resonant frequency, we can keep those Q oscillations in phase, and amplify the response by a factor of Q.

You might think that if some Q is good, more Q is better, but that's not the case for instruments. If we have a vibrating string, we *want* the string to decay, because we want the energy in the vibrating string to turn into sound waves. We can't do that if the string vibrates for a really long time! If you clamp a violin string in a rigid frame, you get Q's of several thousand. If we play a note at a few hundred Hz, then that note will last tens of seconds after we stop playing. While this is not ideal for most music, at least that super-long note will also be super-soft. Put that same string on a violin, which is designed to transfer the vibrational energy of the string into sound waves in the air, and Q drops to several hundred. Now if we play a note, the sound will last for a second, and be much louder.

3. Q and Resonance Width

Another important role Q plays is in the *width* of a resonance. We'll never drive a note at exactly, precisely, 100% the resonant frequency with zero error. That's OK, though. As long as I drive my string in phase with what the string is already doing, I'll amplify the oscillations. That's what we want because we want our instrument to make a noise. As we've seen, if you play two closely-spaced sine waves, you get a beating pattern. The beating happens when the two waves go from in-phase to out-of-phase. For the ideal lossless case, this eventually happens unless the frequencies are identical. If I have damping, though, it doesn't matter if I'm out-of-phase with oscillations that have already damped away. As long as I'm in-phase with the last Q oscillations, I'm still doing a good job of driving the oscillaor. The higher the Q, the closer I need to be to the resonant

frequency.

In practice, we usually drive instruments randomly at a bunch of different frequencies. If we have a resonance at frequency ν with quality factor Q, we would expect our instrument to amplify signals not just at ν but within a frequency range $\delta\nu$ where $\delta\nu \sim \frac{1}{Q}$. Restating, our instrument won't just amplify sounds at ν, it will amplify sounds between $\nu - \nu/Q$ and $\nu + \nu/Q$.

4. Q and Room Resonances

When we talked about the modes of a rectangular room, we saw that the resonant frequencies get more closely spaced as we got up in frequency. How does this affect the acoustics of the room? If the room has a reverb time τ, then the Q of a resonance will be something like T/τ where T is the period. We can swap T for ν since $\nu = 1/T$ to get (up to factors of π):

$$Q \sim \frac{1}{\nu\tau} \tag{4}$$

If we play low notes, the room resonances will be separated by a lot more than the width of the resonance, and we'll hear distinct resonances. If you were designing a concert hall, you might want to damp these out with some Helmholtz oscillators. As we go to higher frequencies, though, the resonances start to overlap with each other. Once that happens, we don't really notice the room resonances and the response will be flat.

5. Discussion

We've seen that damping plays a major role in how oscillators behave. The quality factor Q tells us how many oscillations it takes for our amplitude to decay. Be careful - there are multiple definitions of Q out there, so if you want to use it quantitatively, make sure you know exactly what definition is being used. Some people use the decay in energy over a radian, some the decay over a period, and some the frequency width of the resonance.

For instruments, Q plays a major role in how quickly the sound comes out. If we pluck a string, all the energy we ever have to play with comes from the plucking. If we use a high-Q instrument, we'll hear a less-loud sound for longer. If we use a low-Q one, the sound will come out quickly and will be very loud, but won't last as long.

Q affects rooms as well. The low-frequency modes are often well separated, so we can find the room resonances. At higher frequencies, the modes are more closely spaced, and the resonances can overlap. Once that happens, the room no longer acts like a space with discrete resonances. The reverb time is closely related to Q. Let's take the OSM hall and say it' 40m across and has a reverb time of 2 seconds. Sound moving at 340 m/s will travel 680 metres during 2 seconds, so a sound wave bounces off the walls roughly 20 times during the reverb time. We can use the reverb time to estimate how much sound we lose at each reflection, since $f^{20} = 1/e$. Taking the natural log, we get $20\log(f) = -1$, or $f = \exp(-1/20)$ $sim0.95$. That is, about 95% of the sound energy bounces off the walls at each reflection, and only 5% is lost. If you were in a smaller room, that

had more bounces per second, you would probably want the walls to be better sound reflectors so the reverb time stays around 2 seconds.